Response to Referee Reports

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Preferences for Warning Signal Quality: Experimental Evidence

[Journal Name] · June 2, 2026

We thank both referees for their thorough and constructive reviews. The revised manuscript is stronger for it. Below we address each point in turn. Referee comments are shown in shaded boxes (blue for Referee 1, teal for Referee 2); our responses follow in plain text.

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Associate Editor

The Associate Editor has no additional comments.

Referee 1

We thank Referee 1 for the careful reading of both the original submission and the revision.

R1.1 Incorrect and Inconsistent Definitions of FP and FN Costs

> The paper now emphasizes the distinction between false-positive/false-negative rates and false-positive/false-negative costs. This is a useful clarification. However, the definitions given in the model and in Section 5.2 are not correct. The paper defines FP and FN costs on page 6 as $\pi(1 - P_{01})c$ and $\pi P_{01}L$, respectively, and states that WTP is equally affected by both. There are two problems.
>
> *FP cost:* The expression $\pi(1 - P_{01})c$ is the expected cost of true positive protection, not the cost of false positives. The FP cost should be $(1 - \pi)P_{10}c$, consistent with equation (4) where $\partial b^*/\partial P_{10} = -(1 - \pi)c$.
>
> *FN cost:* The paper defines it as $\pi P_{01}L$ (gross expected loss). An alternative, $\pi P_{01}(L-c)$, captures the incremental cost: under a true-positive signal the agent would have paid c, so the net cost of a miss is $L-c$, consistent with equation (5) where $\partial b^*/\partial P_{01} = -\pi(L-c)$.
>
> This matters for Table 5. The paper should explicitly define the regression variables, map them to equation (3), and verify that the definitions do not affect the reported results.

Response: [Your response here.]

R1.2 Interpretation of Table 5 and Figure 5

> The abstract states that subjects "disproportionately reduce their demand for signals with high false negative rates for rare events." This is imprecise.
>
> Under the risk-neutral null, all Table 5 coefficients should be zero; positive coefficients mean WTP does not fall *enough* as error costs rise. The actual finding is that subjects selectively under-react to the *dominant* cost type:
>
> - Low priors ⇒ FP costs dominate ⇒ subjects fail to adjust for FP costs.
> - High priors ⇒ FN costs dominate ⇒ subjects fail to adjust for FN costs.
>
> Figure 5 corroborates this: empirical sensitivity coefficients are flat across priors whereas theory predicts sharp divergence. The finding is not absolute over-penalisation of FN for rare events but a failure to *differentially* weight FP and FN as the prior shifts. The abstract and conclusion should be rewritten accordingly, and the paper should use "false alarm" and "missed detection" alongside FP/FN notation.

Response: [Your response here.]

R1.3 Mechanism Claim Should Be Softened

> Section 5.3 is much improved, but I remain unconvinced that the data uniquely identify a failure to distinguish FP and FN errors. The same pattern could reflect a total-error-rate heuristic, a "dishonest gremlin" heuristic, probability weighting, noisy Bayesian reasoning, task complexity, or some combination.
>
> A more defensible conclusion: many subjects use a simplified total-error or total-dishonesty heuristic rather than separately processing how FP and FN rates interact with the prior. Relatedly, Section 4 should not be framed only as reassurance that subjects understood the task the BP, IP, and BE patterns are part of the behavioral phenomenon itself.

Response: [Your response here.]

R1.4 Proposition 2 Wording

> Ensure the wording explicitly states that the proposition concerns the ratio of sensitivities $(\partial b/\partial P_{01})/(\partial b/\partial P_{10})$, not absolute levels. A brief economic interpretation would help: risk-averse agents are relatively more intolerant of missed detections but more forgiving of false alarms compared with risk-neutral agents.

Response: [Your response here.]

R1.5 Appendix B: Proof Numbering

> The main text has two propositions but the appendix has three proofs (B.1–B.3). Section B.2 proves a result for imprudent agents that no longer corresponds to any proposition in the main text. B.2 should be dropped or the main text updated to match.

Response: [Your response here.]

R1.6 Appendix B.3: Algebra Error

> In equations (13) and (14) the denominator of the last expression is $(1-\pi)L$; it should be $(1-\pi)c$:
>
> $$\frac{\pi}{1-\pi}\cdot\frac{u'(\xi_1)(L-c)}{u'(\xi_2)\,c}=\frac{\pi(L-c)}{(1-\pi)\,c}\cdot\frac{u'(\xi_1)}{u'(\xi_2)}.$$

Response: We thank the referee for catching this typo. The denominator in equations (13) and (14) has been corrected from $(1-\pi)L$ to $(1-\pi)c$. The stated result is unaffected.

R1.7 Cross-Reference Error

> Page 26 references "Appendix Table A10," but the table with matching content (WTP deviations with IP bias controls) is numbered A8.

Response: [Your response here.]

R1.8 Typos

> - Page 7: "Equations 6-s 7" → "Equations 6–7."
> - Page 15, fn. 10: missing comma between $\hat{\pi} = 0.43$ and $\hat{\theta} = 0.25$.
> - Page 18: "the value for for a prior of 0.2" duplicated word.
> - Page 27: "the observed sensitivity patterns emerges" → "emerge."
> - Page 27: "characteristcs" → "characteristics."

Response: All five typos have been corrected. We thank the referee for the careful reading.

Referee 2

We thank Referee 2 for the detailed and constructive comments on the revision.

R2.1 Treatments and Treatment Assignment

> **(a)** The "Treatments and Treatment Assignment" subsection states that each subject is randomly allocated to a pair of priors but does not explain how information structures are assigned.
>
> **(b)** The text notes that "the first treatment was always the one with a fully truthful information structure," suggesting information structures are *not* randomly assigned. The authors should provide a general description and justification.
>
> **(c)** Because information structures do not appear to be randomly assigned, sample sizes across treatment cells (prior $\times$ information structure) likely differ. The authors should report sample sizes by treatment cell, e.g. in an appendix table ($4 \times 12 = 48$ cells).

Response: [Your response here.]

R2.2 Description and Groupings in Tables 2, 3, and A4

> **(a)** Page 12 states that "Table 2 presents the average protection decisions by prior and signal type," but Table 2 actually groups observations by signal value (white/black) and information-structure type (truthful, FP-only, FN-only, both errors). The same mismatch appears in the caption of Table A4.
>
> **(b)** The authors could briefly justify why Tables 2, 3, and A4 pool across priors. A single sentence would suffice.

Response: [Your response here.]

R2.3 Claim on Page 16 Regarding Introduction of the FN Rate

> **(a)** The paragraph below Table 3 claims "a similar effect for the introduction of the FN rate (row 1 vs. 2)," but the mean updating error barely changes (0.048 vs. 0.051).
>
> **(b)** Moreover, introducing the FN rate changes the Bayesian posterior for the white signal (from 0.000 to 0.119), so the row 1 vs. row 2 comparison does not hold the posterior fixed unlike row 1 vs. row 3.

Response: [Your response here.]

R2.4 Section 4 Closing Summary Understates the Evidence

> The paragraph before Section 5 concludes only that "subjects understand the experimental environment well enough." But Tables 2 and 3 also show that the FP (FN) rate

affects protection decisions and updating errors even when it should not influence the Bayesian posterior suggestive evidence for the FP/FN-confusion mechanism. A sentence foreshadowing this in Section 4 would make the main mechanism visible earlier.

Response: [Your response here.]

R2.5 Hypothesis 1 and Result 1 Overstate the t-Test Evidence

Hypothesis 1 states WTPs "are equal to their value for risk-neutral agents," and Result 1 concludes "no significant discrepancies." But the t-test only establishes that $E[\Delta b_{is}] = 0$; large but offsetting deviations could cancel. Suggested revisions:

i. Restate more precisely: "WTPs are not systematically above or below the risk-neutral benchmark *on average*"; or

ii. Test the stronger claim by examining whether $|\Delta b_{is}|$ is significantly different from zero.

Response: [Your response here.]

R2.6 (Optional) Reorganise Section 5.3 to Lead with FP/FN Confusion

Given that FP/FN confusion is the primary mechanism, it might be preferable to present it first and then evaluate and rule out alternative explanations.

Response: [Your response here.]

R2.7 Risk-Preference Discussion (pp. 23–24)

The authors could note that the main effects of FP and FN costs remain significant, rather than focusing solely on the insignificance of the interaction terms.

Response: [Your response here.]

R2.8 Probability Weighting Discussion (pp. 24–25)

(a) In equation (9) and the definition of Λ, the numerator should be P_{01} rather than P_{00}.

(b) The argument with rank-dependent utility is appreciated. The flow is clear and the observation that $\pi + (1 - \pi)P_{10} \leq 0.75$ is insightful. However, this depends on the rotation point of the weighting function. If the rotation point lies far to the left, that range might still fall in the steeper region, implying lower relative sensitivity. Evidence for "middle" rotation points (e.g. Gonzalez and Wu, 1999) would be helpful, though the flexibility of the weighting function may make this difficult to rule out entirely.

Response: [Your response here.]

R2.9 Anticipation of Bias in Protection Decisions

(a) Table A10 should be Table A8; the caption of Table A8 should refer to WTP *deviations* rather than simply WTP.

(b) The two new variables (Δprot, Δloss) may not adequately test the "anticipation of bias" mechanism. Negative coefficients suggest that subjects who perform worse in the IP task are *more* accurate in WTP, which could indicate they anticipate their own mistakes.

(c) Given that the coefficients are statistically significant, concluding that "biases in IP do not explain the observed pattern" seems premature. A more appropriate interpretation is that the coefficients on FP and FN do not *change* after introducing these variables.

(d) A more informative specification would include interaction terms between mistake variables and FP/FN rates. If anticipation of bias is the primary channel, mistakes should mediate the WTP–FP/FN relationship.

(e) Even if anticipation of bias is important, it does not undermine FP/FN confusion; Tables 2 and 7 already show that IP decisions respond to FP/FN in ways consistent with confusion. Anticipation of bias would reinforce, not replace, the FP/FN-confusion mechanism.

Response: [Your response here.]

R2.10 Cross-Task Consistency Analysis

The authors should consider incorporating the cross-task consistency analysis from the response letter (Table 2, page 13) into the paper or at least the appendix.

Response: [Your response here.]

R2.11 (Optional) "Dishonest Gremlin" Framing

Even without the "dishonest" framing, individuals may still conflate FP and FN errors. A brief discussion of this potential limitation (perhaps at the end of the paper) would be helpful.

Response: [Your response here.]

R2.12 Exposition

(a) Page 11: "this section also provides ... the theoretical predictions" is imprecise the predictions are those of the risk-neutral benchmark specifically.

(b) Relabel "Posterior" as "Bayesian posterior" in Tables 2, 3, A2, and A4.

(c) In Tables 5 and A8, a horizontal rule separating the main regression panel from the risk-averse/risk-loving subgroup tests would improve readability.

(d) (Optional.) Adding a short closing sentence to each appendix table caption pointing to the key coefficient or column would reduce cross-referencing costs.

(e) Page 21: "the systematic WTP biases reported above" is vague please specify which biases are meant.

Response: [Your response here.]

R2.13 Typos

- Equation (2), page 5: extra "=" sign.
- Page 7: "Equations 6-s 7" → "Equations 6 and 7."
- Footnotes 10 and 11: missing full stop at end.
- Footnote 13, page 21: "identical" should likely be "different."
- Last paragraph, page 26: "Table A10" → "Table A8."

Response: All five typos have been corrected. We thank the referee for the careful reading.